

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

14th September 2017

Subject CT4 – Models

Time allowed: Three Hours (10.30 – 13.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. Attempt all questions, beginning your answer to each question on a separate sheet.*
- 3. Mark allocations are shown in brackets.*
- 4. Please check if you have received complete Question paper and no page is missing. If so, kindly get a new set of Question paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer booklet and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1) i)** List eight factors which should be considered when assessing whether a model is suitable for a particular application. (4)
- ii)** State, giving reasons, a factor which would be particularly important in each of the following applications:
- Calculating the pension contribution for a medium sized pension scheme.
 - Working out how much it will cost to buy each member of your team their favourite cake on your birthday in six months' time. (2)
- [6]

- Q. 2)** A newly appointed actuary of a large insurance company feels that current model of mortality used for mortality investigation and assumption setting is rendered obsolete due to change in policyholders mix, behavioural change in buying insurance in general population and technological & medical advancements. He wants to assess the suitability of various models to select the one suitable for mortality investigation for this company. Binomial model is one of the models considered for the same. However, the actuary believes that it may be complex to build and comprehend. There is statistical software to implement this model easily but he wants a simplified model to understand the model clearly using a simplified data. He has asked one of his juniors to build a prototype and show him the calculations and has specifically asked for demonstration of some key calculations mentioned in part (i) to (iv). The junior sampled the following data from a previous mortality investigation carried out sometime back.

Data of Several lives of age $x = 50$ Last birthday at 01/03/2017 that were observed from age $x+a$ to age $x+b$ is provided to him in the table below.

Life i	Date of entry	Investigation end date	Date of exit	Mode of exit
1	01/01/2016	01/01/2017	01/01/2017	
2	01/01/2016	01/10/2016	01/07/2016	Death
3	01/04/2016	01/01/2017	01/10/2016	Death

The investigation end date is the 51st birthday of each life.

- i)** List the strengths and weaknesses of the Binomial model for the estimation of empirical mortality rates. (2)
- ii)** Write down the likelihood function for the whole group based on binomial model. (2)
- iii)** Calculate the likelihood for the group assuming the Balducci's assumption in terms of q_x ?
- [Note: Balducci's assumption is given ${}_{b-a}q_{x+a} = \frac{(b-a)q_x}{1 - (1-b)q_x}$ as: (2)
- iv)** Find the MLE of q_x based on the likelihood function in (iii) above? (3)
- [9]

Q. 3) An exercise to recruit new staff is conducted by a large company. The company wants to optimize the entire recruitment process. Following is the statistics related to the process aggregated over last few years.

-The first stage is the arrival of candidate where candidates have to wait in queue to fill application forms on the kiosks set up for the purpose. The process takes on average α hours.

- Further an online screening test is conducted which takes on average β hours.

- Around $\rho_1\%$ of initial total applicants does not qualify this test. The qualified applicants are further bifurcated for technical and non-technical roles depending on the score in the test and their qualifications.

- $\rho_2\%$ of initial total candidates is selected for technical roles and are given brief of their roles which takes around γ hours.

- $\rho_3\%$ of initial total candidates is selected for non-technical roles and are given brief of their roles which takes around δ hours.

It may be assumed that company is very big and there are always vacancies at all levels and therefore recruitment process is continuously carried out. Also, there is no shortage of candidates.

A continuous time Markov model is proposed to model the operation of the recruitment process with states being A (Application form), S (Screening test), T (Technical roles), N (Non-Technical roles) and E (Process ends).

i) State the distribution followed by time spent in each state. (1)

ii) Write the generator matrix for this process. (2)

iii) Write the forward differential equations for p_{AS} and p_{AT} (3)

iv) Derive an expression for p_{AA} (2)

v) Hence derive expression for p_{AS} (3)

[11]

Q. 4) An investigation was undertaken into the mortality of patients suffering from a particular infection outbreak in a town who were treated in a hospital. A sample of patients was observed from the date of admission until they died or left the hospital or a period of 45 days elapsed. The following data relate to 13 patients treated in a hospital for infection.

Patient Number	Duration of Observation (days)	Reason for observation ceasing
1	5	Died
2	7	Died
3	14	Died
4	18	Discharged
5	25	Discharged
6	28	Died
7	30	Discharged

8	34	Discharged
9	35	Died
10	36	Discharged
11	45	Observation ended
12	45	Observation ended
13	45	Observation ended

- i) State the types of censoring that are present in the above investigation and explain where they occur? (2)
- ii) Comments on whether censoring likely to be informative in the investigation? (1)
- iii) Calculate the Kaplan-Meier estimate of the survivor function for the patients. State any assumptions that you have made. (4)

At the end of investigation, the hospital claims in a press conference that more than 80% of patients who responded to treatment for the initial two weeks are recovered from the infection and likely to be discharged from hospital.

- iv) Comment on the hospital's claim in view of the above investigation? (3)
- [10]

Q. 5) A company is studying the health records of its longest serving employee in order to improve its provision for health insurance. Let $X(t) = H$ if the employee is healthy at time t , $X(t) = S$ otherwise. The available information includes the value of $X(t)$ for all $0 \leq t \leq T$

- i) Explain the principal stages in the formulation and verification of a stochastic model for this process. (3)
- ii) The company chooses to model X as a two-state time-homogeneous Markov jump process with transition rates $\sigma_{HS} = \sigma, \sigma_{SH} = \rho$
- a) Write down estimates for the parameters σ and ρ of the model in terms of quantities which may be derived from the available data. (2)
- b) Indicate one test which could be used to determine whether the data support the assumption that the Markov jump process model is suitable. (2)

[7]

Q. 6) An investigation was undertaken for the mortality of a group of policyholders who purchased policies of an endowment plan, which was introduced couple of years back. An extract of the experience data of the company for this plan is given below:

	Number of policies in force as at the financial year closing date using age last birthday		
Age last birthday	31-03-2014	31-03-2015	31-03-2016
40	1220	1430	1510
41	1210	1350	1470
42	1140	1350	1510

Further, total 14 deaths at age 41 year, classified by age last birthday, were reported to the company in the Calendar year 2015 who had taken this plan.

- i) State the principle of correspondence as it applies to the estimation of mortality rates. Explain why it may be difficult to ensure the principle of correspondence is adhered to? (2)
 - ii) Estimate the death rate (force of mortality) at age 41(last birthday) in the calendar year 2015 stating all the assumptions. (3)
 - iii) List the data required for exact calculation of central exposed to risk for lives aged x last birthday for a mortality investigation over the three year period from 31.03.2014 to 31.03.2017. (2)
- [7]**

Q. 7) Customers in a certain city are continually switching the brand of toothpaste they buy. If a customer is now using brand A, the probability he will use brand A next week is 0.5, that he switches to brand B is 0.2 and that he switches to brand C is 0.3. If he now uses brand B, the probability he uses B next week is 0.6 and the probability that he switches to C is 0.4. If he now uses brand C, the probability he uses C next week is 0.4, that he switches to A is 0.2 and to B is 0.4. Assume the process is a Markov Chain.

- i) Determine the transition matrix for this process and check if transition probabilities are consistent. (2)
 - ii) Find the probability of customer using brand B in two weeks' time if he is currently using brand A. (1)
 - iii) If the percentage of customers are now using brand A is 30%, the percentage using brand B is 20% and the percentage using brand C is 50%, find the percentage of customers using brand C in three weeks' time. (2)
 - iv) Find the probability of customer using brand B in six weeks' time if he is currently using brand A. (2)
 - v) Commenting on irreducibility and periodicity of this chain, find the percentage of customers using each brand of toothpaste in the long run. (3)
- [10]**

Q. 8) An investigation was carried out for mortality of people aged between 50 and 60 years collecting data on the following covariates for each person:

X_1 Sex (a categorical variable with 0 = female, 1 = male)

X_2 Cigarette smoking (a categorical variable with 0 = non-smoker, 1 = smoker)

- i) State the Gompertz Law? (1)

In addition, data were collected on the age at death for persons who died during the period of investigation.

In order to analyze the above data, it was decided to use a Gompertz hazard, $\lambda_x = Bc^x$; where age (x) is assumed as the duration since the start of the observation.

- ii) Show that the substitution:

$$B = \exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2);$$

in the Gompertz model (where $\beta_0 \dots \beta_2$ are parameters to be estimated), leads to a proportional hazards model for this particular analysis. (3)

Using the Gompertz hazard, the parameter estimates in the proportional hazards model were as follows:

Covariate	Sex	Smoker
Parameter estimate	β_1	β_2
Parameter	+0.30	+0.65

The parameter value of β_0 is -4.00 and c is +1.05 .

- iii) Describe the characteristics of the person to whom the baseline hazard applies in this model. (1)
- iv) Calculate the estimated hazard for a female cigarette smoker aged 54 years. (2)
- v) Show that, according to this model, a cigarette smoker at any age has a risk of death roughly equal to that of a non-smoker aged thirteen years older. (4)

[11]

Q. 9) A fair coin is tossed repeatedly, with the tosses being independent. Define a Markov chain with states S,H,T,HT, where S is a starting state, H indicates a head on the current toss, T indicates a tail on the current toss(without heads on the previous toss), and HT indicates heads followed by tails over the last two tosses.

- i) Draw the diagram showing all states mentioned above along with transition probabilities. (2)
- ii) Write the transition matrix and determine the probability of observing head followed by tail during 5th toss. (2)
- iii) Determine the periodicity and state whether process is irreducible. Also comment on whether the process is stationary. (2)
- iv) What is the expected number of tosses necessary to first observe a head directly followed by tails? (2)
- v) During the experiment, it is observed that head is followed by a tail, what is the expected number of additional tosses until we again observe a head followed directly by a tail? (2)
- vi) Set up another similar Markov chain from which we could calculate the expected number of tosses necessary to first observe tails directly followed by tails. (2)
- vii) What is the expected number of tosses necessary to first observe a tail directly followed by a tail? (1)

viii) A tail followed by a tail is observed, what is the expected number of additional tosses until a tail followed directly by a tail is observed?
Note that the number of additional tosses could be as little as one, if tails were to come up again. (1)

ix) Comment on the results obtained above. (2)
[16]

Q. 10) i) List the methods of graduation suggesting a situation for each method in which its use may be appropriate. (3)

A life insurance company is aggressively writing protection business targeting young lives. An investigation into the mortality of young professionals engaged in IT industry was carried out. The following table gives an extract of data from the results.

Age	Initial exposed to risk	Observed Deaths	Graduated Rates ($q'x$)
20	1060	12	0.0125
21	1250	14	0.0118
22	1210	16	0.0134
23	700	9	0.0102
24	875	11	0.0111
25	950	15	0.0137
26	805	10	0.0126
27	1390	16	0.0127
28	1080	17	0.0122
29	1310	14	0.0131

ii) Carry out Chi-square test for overall goodness of fit on the data, using a 95% significance level and appropriate degrees of freedom. (4)

It has now been decided to produce graduated rates multiplied with a constant factor across all the ages. The constant factor is derived as a ratio of total number of actual deaths to the total number of expected deaths for the given data.

iii) State how you would expect the results to differ, if at all, from the results of the test carried out in part (ii) above? (3)

iv) Describe three shortcomings of the χ^2 test for comparing crude estimates of mortality with a standard table and why they may occur. Name a test that would detect each of those shortcomings. (3)
[13]
